Quantitative Methods – Human Sciences

Potential outcomes and experimental design

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0.1 How to use mathematical notation

You can insert in-line equations using LaTeX~ notation like this: $y_i = \alpha + \beta_\tau \tau + \epsilon_i$ and they will show like this: $y_i = \alpha + \beta_\tau \tau + \epsilon_i$. If you want equations to be separated from the main body, just wrap them around double dollar signs instead of one, like this: $$y_i = \alpha + \beta_\tau \tau + \epsilon_i$$ for the following result:

$$y_i = \alpha + \beta_\tau \tau + \epsilon_i$$

0.2 Simulate a Data Generating Process (DGP)

To insert R code (that will be rendered directly in this document), use the following syntax:

```
# This is R code
# Code and results will be shown in the compiled document

# Set seed for reproducibility
set.seed(1492)

# Let's made a simple calculation
# assuming the following model
# Y0 ~ Norm(10,2)
# Y1 = Y0 + \tau + epsilon_i
# epsilon ~ Norm(0,1)
# tau = 1.5

n <- 1000 # we will create 1K cases
tau <- 1.5 # a treatment effect that is fixed for everyone
epsilon <- rnorm(n) # some "random noise"
y0 <- rnorm(n, mean = 10, sd = 2) # control PO, from normal distribution
y1 <- y0 + tau + epsilon # treated PO, control + treatment effect + noise

# Now, let's create one possible random assignment vector
```

```
# for a binary treatment with p=0.5
d <- rbinom(n, 1, 0.5)

# Let's create the observed outcome variable, using the switching equation
y <- y1*d+(1-d)*y0
```

0.3 Plots and figures

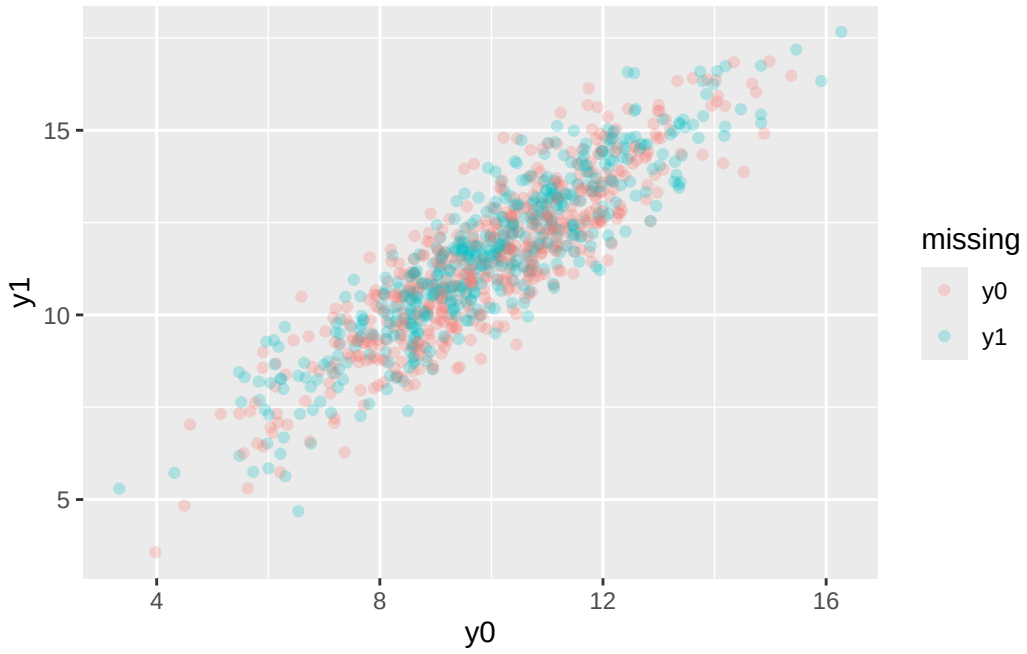
We can plot the data to explore what is in there. Plots will be rendered in the final document if you include their code in this document:

```
# Simple plot example
# First, let's combine our data in a single data.frame

data <- data.frame(y1,y0,y,d)
data$missing <- ifelse(d==1, "y0","y1")

library(ggplot2) # Load ggplot2 package

# Create a scatterplot, highlighting observed vs missing outcomes
ggplot(data, aes(x = y0, y = y1, color = missing)) +
 geom_point(alpha = 0.25)
```



0.4 ATE estimation

```
# Now, let's calculate the ATE
# These two ways are equivalent (why?)

mean(y1-y0)
```

[1] 1.560768

```
mean(y1) - mean(y0)
```

[1] 1.560768

```
# Now, let's approximate the ATE with a difference-in-means
# Think: why cannot we simply use mean(y1-y0 anymore?)
# Think: why does this ATEhat differs from the ATE calculated above?
# Think: is it still a good approximation?

mean(y[d==1]) - mean(y[d==0])
```

[1] 1.439973

1 Now it's your turn

(No worries for now, we will populate this section together in class)

1.1 Randomisation inference

1.2 Simple randomisation

1.3 Block randomisation

1.4 Clustered randomisation